Taminul Islam

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Summary

Machine Learning PhD researcher specializing in efficient LLMs/VLMs, multimodal multi-task learning, and multi-agent systems with 31 peer-reviewed publications and 500+ citations. Built production-ready transformer models achieving 99.3% accuracy on 203K+ image datasets, with 2 papers accepted to ICCV 2025 (25% acceptance rate). Proven track record delivering lightweight AI solutions (1.38M-5M parameters) outperforming state-of-the-art baselines while optimizing inference speed and memory efficiency. Eligible for F-1 OPT/CPT internships.

Technical Skills

Languages & Frameworks: Python, TensorFlow, PyTorch, Keras, NumPy, Scikit-learn, LoRA/QLoRA, Quantization, JavaScript

Computer Vision & Multimodal AI: Semantic Segmentation, Vision Transformers (ViT, Swin), VLM, SAM-2, YOLO, OpenCV

ML Specializations: Efficient LLMs/VLMs, Multi-Task Learning, Multi-Agent Systems, Model Optimization, Parameter-Efficient Fine-Tuning

Data & Analytics: SQL (PostgreSQL, MySQL), Power BI, Tableau, Excel, Data Visualization, Statistical Analysis

Tools & Platforms: Git/GitHub, Google Cloud Platform, Gemini API, Android Debug Bridge (ADB), Docker, LATEX

Education

Southern Illinois University Carbondale, PhD in Computer Science — Jan 2024 – Present

- GPA: 4.0/4.0 | Graduate Research Assistant in AI/ML/Computer Vision at BASE Lab — Carbondale, IL

Daffodil International University, BS in Computer Science and Engineering — Jan 2018 – Dec 2021

- GPA: 3.52/4.0 | Full-Free Scholarship (Top 2% of applicants) — Dhaka, Bangladesh

Experience

Research Assistant – AI/ML/Computer Vision, BASE Lab, SIUC — Jan 2024 – Present

- Architected CarboFormer, a lightweight semantic segmentation model (5.07M parameters) achieving 84.88% mIoU for CO2 emission quantification—15% improvement over baseline with 3x faster inference speed; accepted to ISVC 2025 (Oral presentation) [github.com/taminulislam/carboformer]
- Co-developed GasTwinFormer, a hybrid vision transformer for livestock methane detection, processing optical gas imaging data for dietary classification; achieved 92% segmentation accuracy and accepted to ICCV 2025 (Oral, top 1.5% of submissions) [github.com/toqitahamid/gastwinformer]
- Built WeedSwin hierarchical vision transformer achieving 99.3% accuracy in multi-stage weed detection across 203,567 images; published in Scientific Reports (Nature, Q1 journal) and featured in WeedSense multi-task architecture for ICCV 2025 (Poster) [github.com/taminulislam/weedswin]

Executive, ServicEngineBPO Ltd. — Aug 2022 – Nov 2023

- Analyzed system performance data using SQL and Excel to optimize IT infrastructure for 50+ users; built automated reporting dashboards tracking digital advertising metrics, reducing system downtime by 40% through data-driven decision-making

Research And Development Assistant, Quadque Technologies Ltd. — Feb 2022 – Jun 2022

- Built predictive analytics models using Python, SQL, and Scikit-learn for client applications; performed feature engineering, and statistical analysis; created Tableau dashboards communicating insights to business stakeholders

Undergraduate Research Assistant – Machine Learning, Daffodil International University — Apr 2020 – Jan 2022

- Published 15+ peer-reviewed papers on cybersecurity, NLP, and medical imaging using TensorFlow, PyTorch, and NumPy; developed breast cancer prediction model with 97% accuracy using XGBoost on 500-patient dataset (Scientific Reports, 55+ citations)
- Engineered deep learning pipelines for disaster tweet classification, cardiovascular disease prediction, and fake news detection, achieving 90%+ accuracy across all domains with optimized hyperparameter tuning

Projects

Mobile QA Multi-Agent System | *Python, Gemini API, ADB, Google ADK* [github.com/taminulislam/mobile-qa-agent]

- Engineered multi-agent testing framework with Supervisor-Planner-Executor architecture for automated mobile QA; integrated vision-language model (Gemini 2.5 Flash) for screenshot interpretation and natural language test case execution on Android emulator
- Implemented differentiated failure detection system achieving accurate reporting of execution vs. assertion failures; demonstrated modular LLM integration enabling seamless substitution of language models across agent roles

EfficientMTNet: Lightweight Brain Tumor Multi-Task Network | *PyTorch, Deep Supervision* |
[github.com/taminulislam/EfficientMTNet]

- Developed ultra-compact multi-task learning architecture (1.38M parameters) for simultaneous brain tumor classification and segmentation, achieving 99.07% accuracy, 76.25% mIoU, and 1124.5 FPS inference speed—98.5% smaller than Vision Transformer with highest normalized efficiency score (1.000); deployed deep supervision strategy for enhanced gradient flow in resource-constrained medical imaging

MESA: Medical LLM Safety & Accuracy Evaluation | *PyTorch, LoRA/QLoRA, 4-bit Quantization* |
[github.com/taminulislam/GenAI_MESA]

- Evaluated 3 fine-tuned medical LLMs (117M-2.7B parameters) across 20,000 samples using 16 metrics spanning accuracy, safety, and efficiency; applied parameter-efficient fine-tuning (LoRA) training only 0.1-1% of parameters with 4-bit quantization for 75% memory reduction
- Optimized BioGPT-Large achieving 67.2% exact match accuracy and 78.1% medical entity recognition for clinical deployment; demonstrated DialoGPT-Small 100% safety rate with 0.22 GB GPU memory for patient-facing applications on RTX 3090

Selected Publications (32 total, 500+ citations, h-index: 14)

- **Islam, T.**, et al. (2025). "CarboFormer: Lightweight Semantic Segmentation for CO2 Detection." *ISVC 2025* (Oral)
- **Islam, T.**, et al. (2025). "WeedSwin: Hierarchical Vision Transformer with SAM-2 for Weed Detection." *Scientific Reports* 15(1), 23274
- **Islam, T.**, et al. (2024). "Breast Cancer Classification with Explainable AI." *Scientific Reports* 14(1), 8487

Awards & Leadership

Honors: 3rd Place, Illinois Young Innovator of the Year (Top 3 of 15 finalists, Falling Walls Lab Illinois 2025) | Associate Editor, Frontiers in Medicine (Q1 journal, youngest appointee) | ACM Professional Member | Reviewer, Scientific Reports

Leadership: General Secretary, Bangladesh Student Association (secured Best RSO Award 2025 for 150+ member organization) | Captain, Cricket & Badminton Championship Teams (2024-2025)